

Community Assembly

Today's Agenda:

- Network assignment reminder
 - Paper presentation 2
 - Quiz
 - Community Assembly

Network assignment

Reminder: Network assignment is due Friday (14 March) by midnight

Paper presentation 2: Community Assembly (26 March)

Step 1: Find a paper on community assembly in your system (or one you pick)

Step 2: Make 2 slides explaining the paper

Slide 1) graphics made by you that explain what was done in the study

Slide 2) what they found (should include results figure(s) from paper)

Step 3: Present the paper using those two slides (5 minute max)

Paper presentation 2: Community Assembly (26 March)

Grade Breakdown:

- Graphics made by student help convey key information 20%
- Minimal text 20%
- Overall slide quality 10%
- Understanding and explanation of the chosen paper 20%
- Focus on key findings 20%
- Overall presentation quality 10%

Questions on assignments?

Community assembly

Why do these particular species occur in this community?

- Essentially the core question of community assembly theory

Community assembly

Community assembly focuses on a few key concepts:

- 1) Regional species pool

Community assembly

Community assembly focuses on a few key concepts:

- 1) Regional species pool
- 2) Dispersal

Community assembly

Community assembly focuses on a few key concepts:

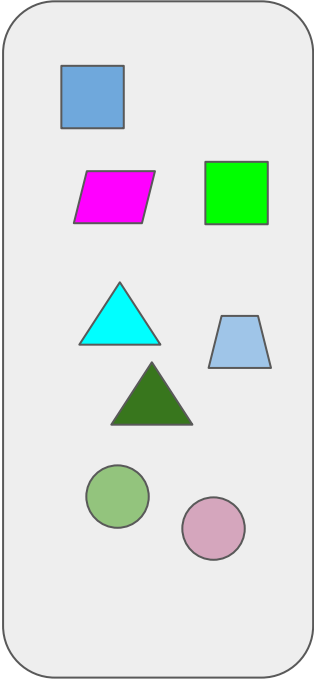
- 1) Regional species pool
- 2) Dispersal
- 3) Environmental filtering

Community assembly

Community assembly focuses on a few key concepts:

- 1) Regional species pool
- 2) Dispersal
- 3) Environmental filtering
- 4) Biotic filtering

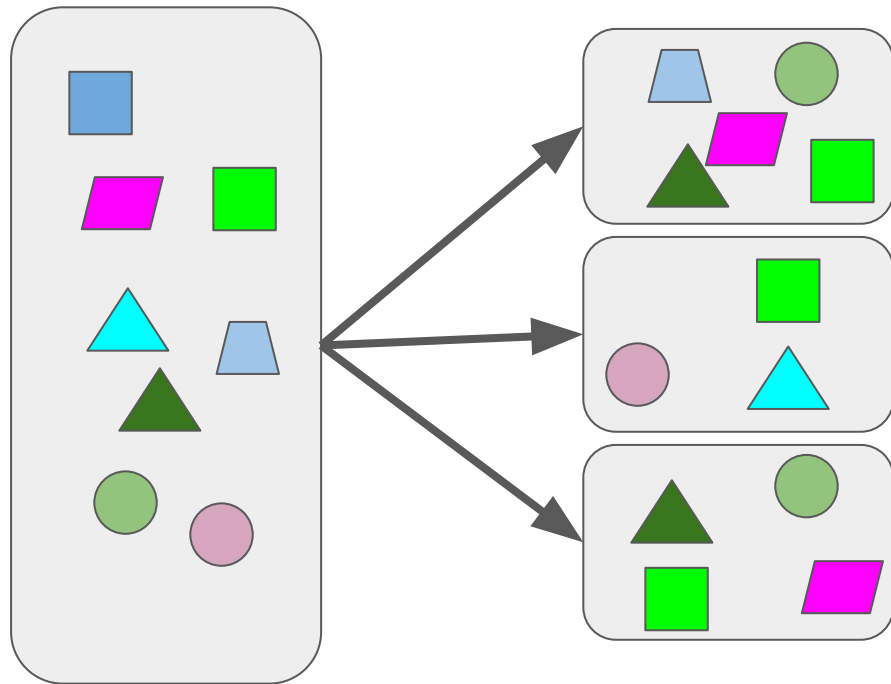
Species pool

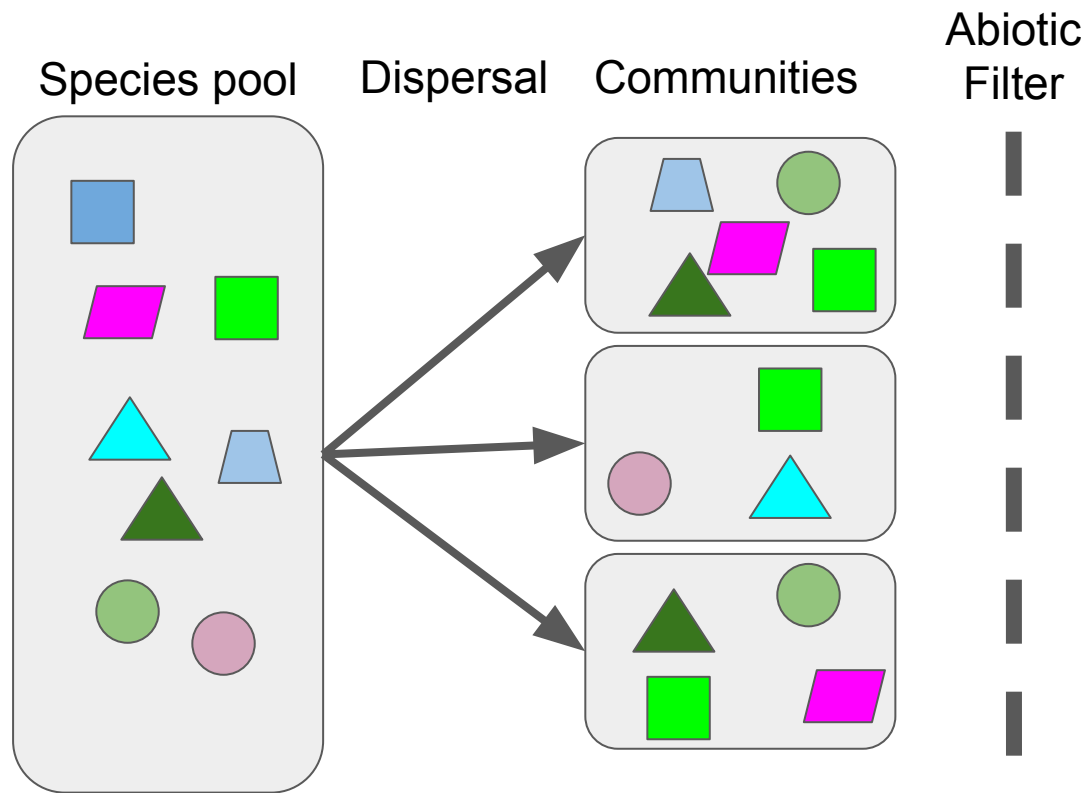


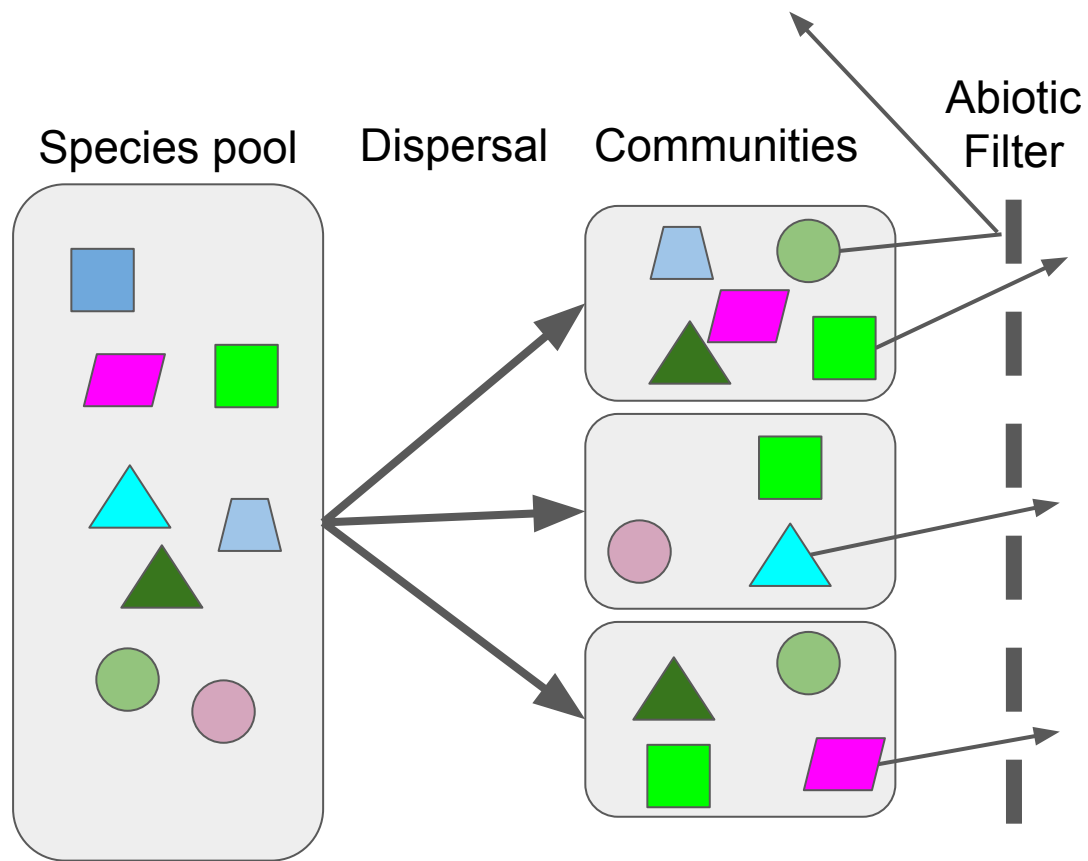
Species pool

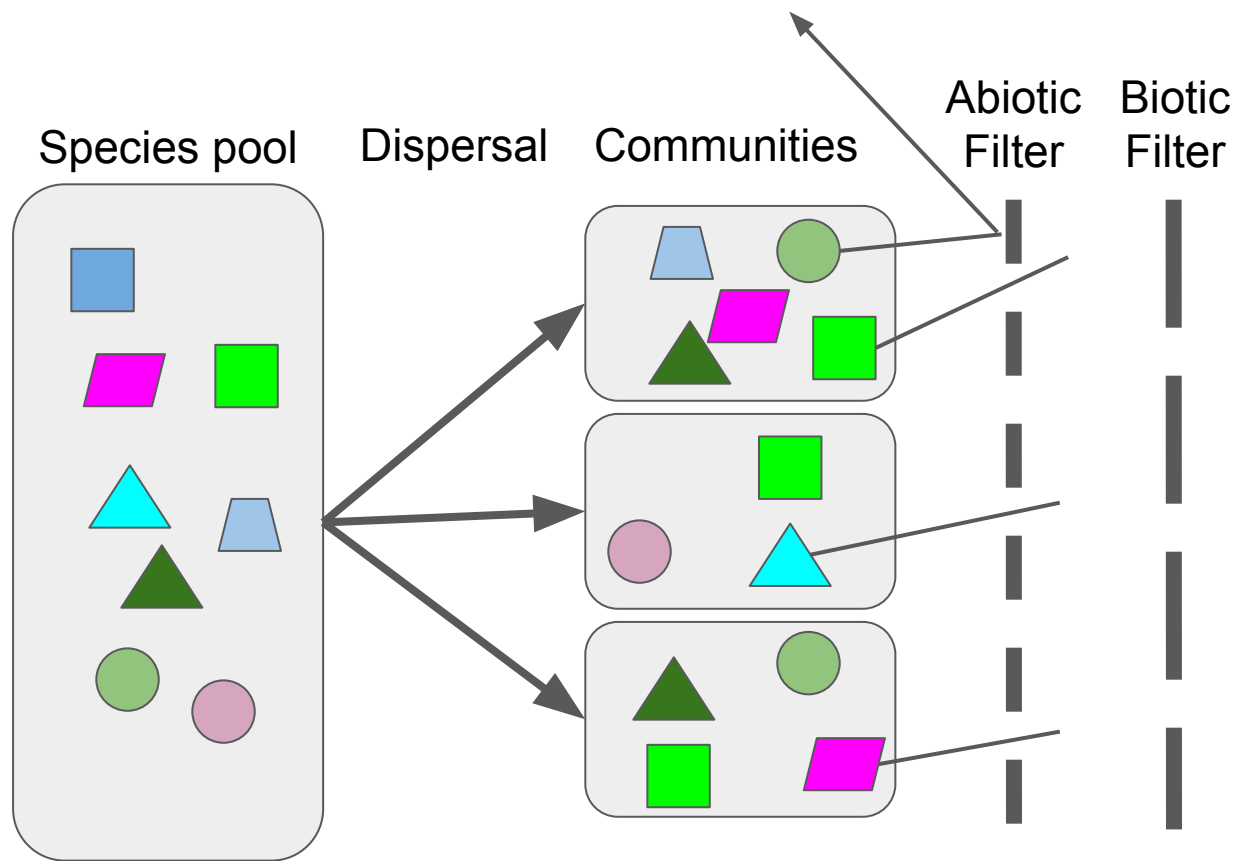
Dispersal

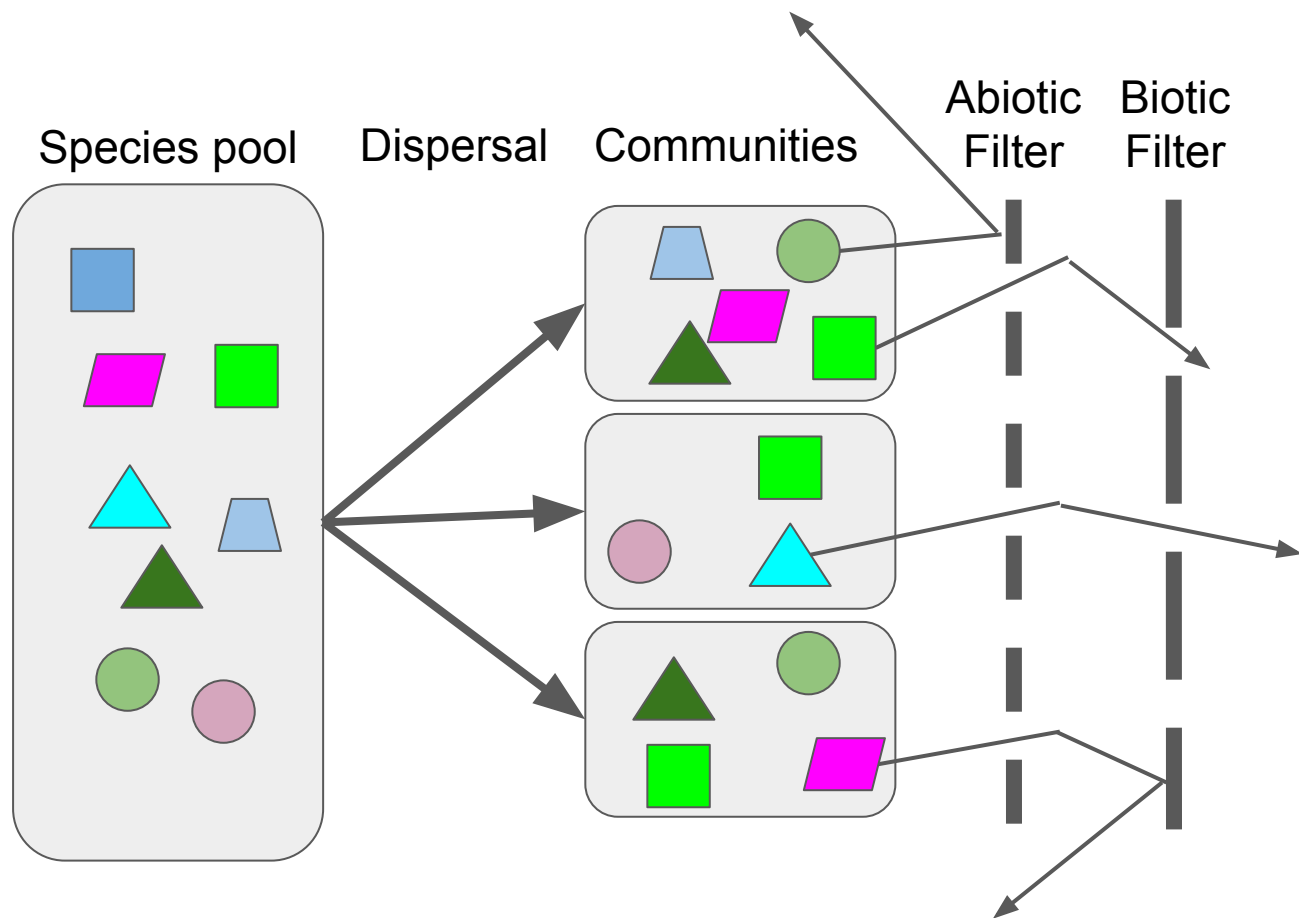
Communities

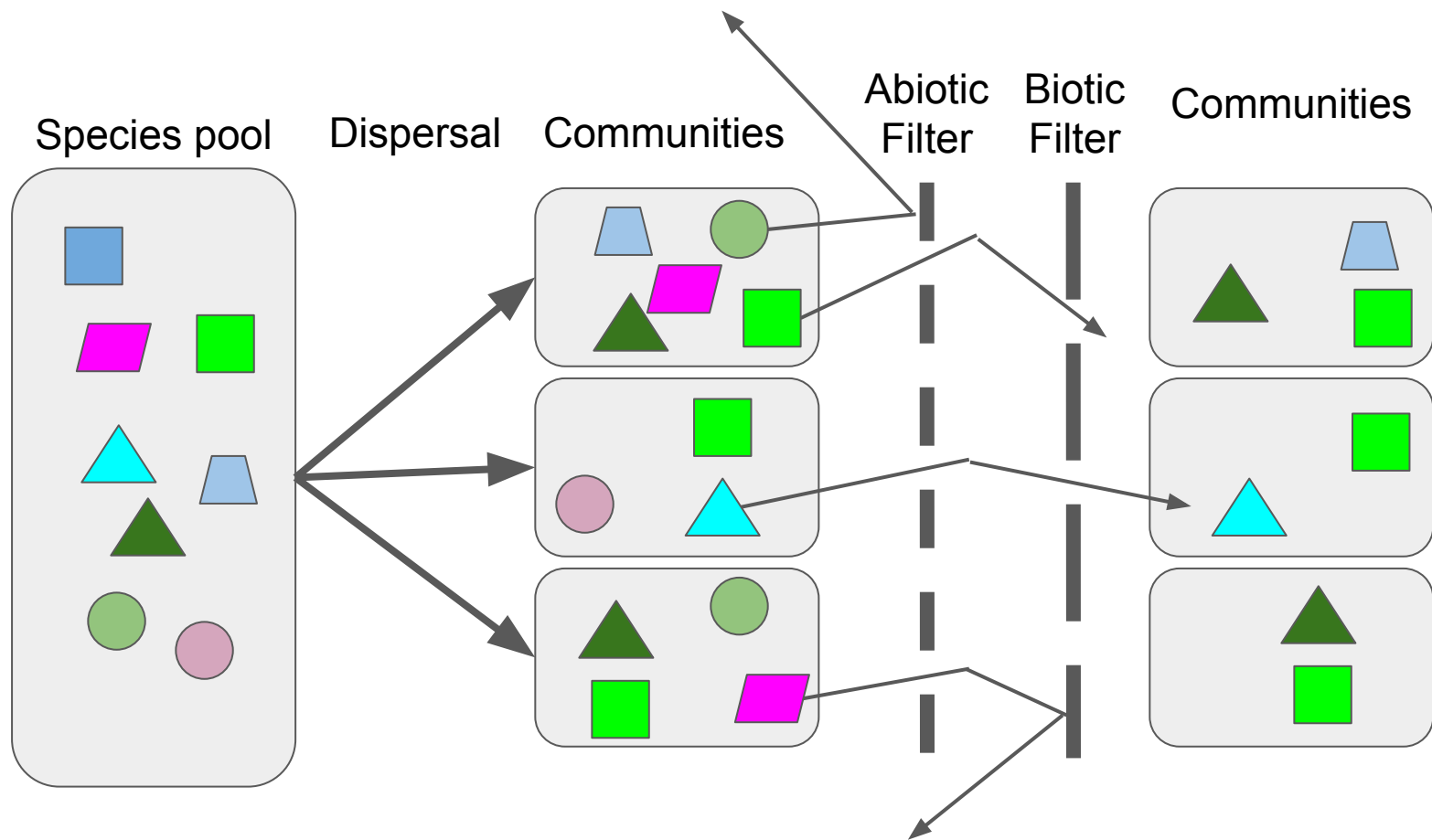


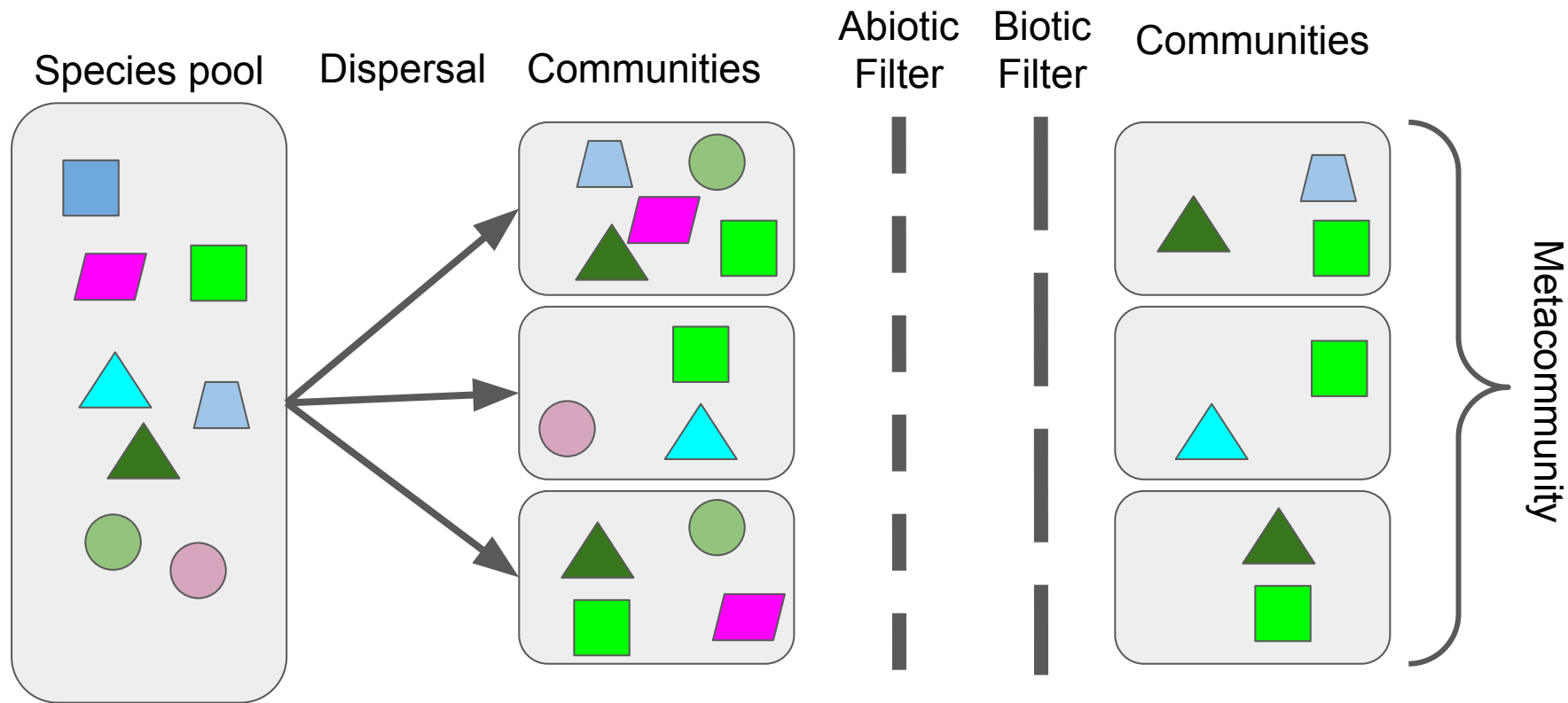


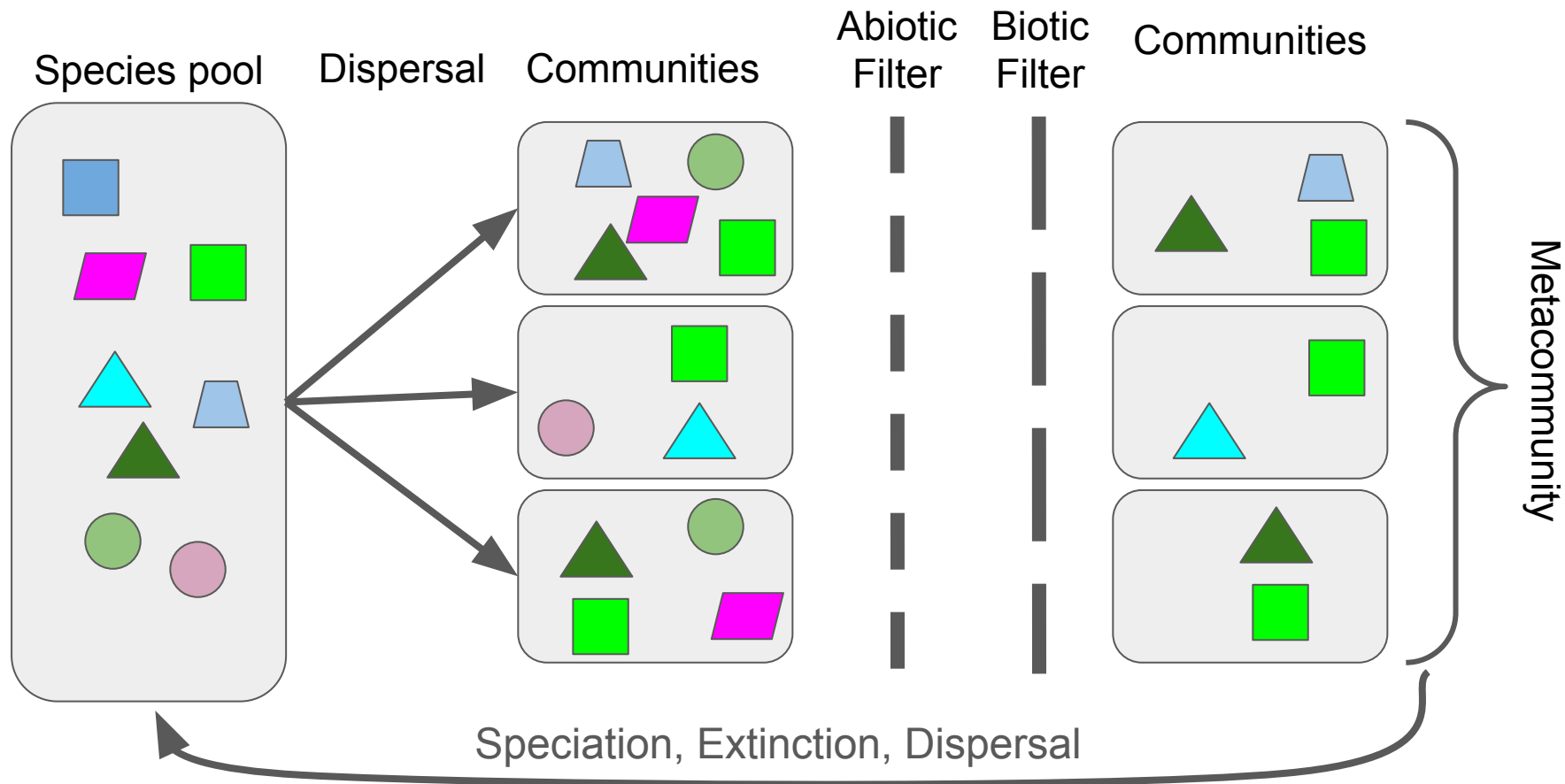


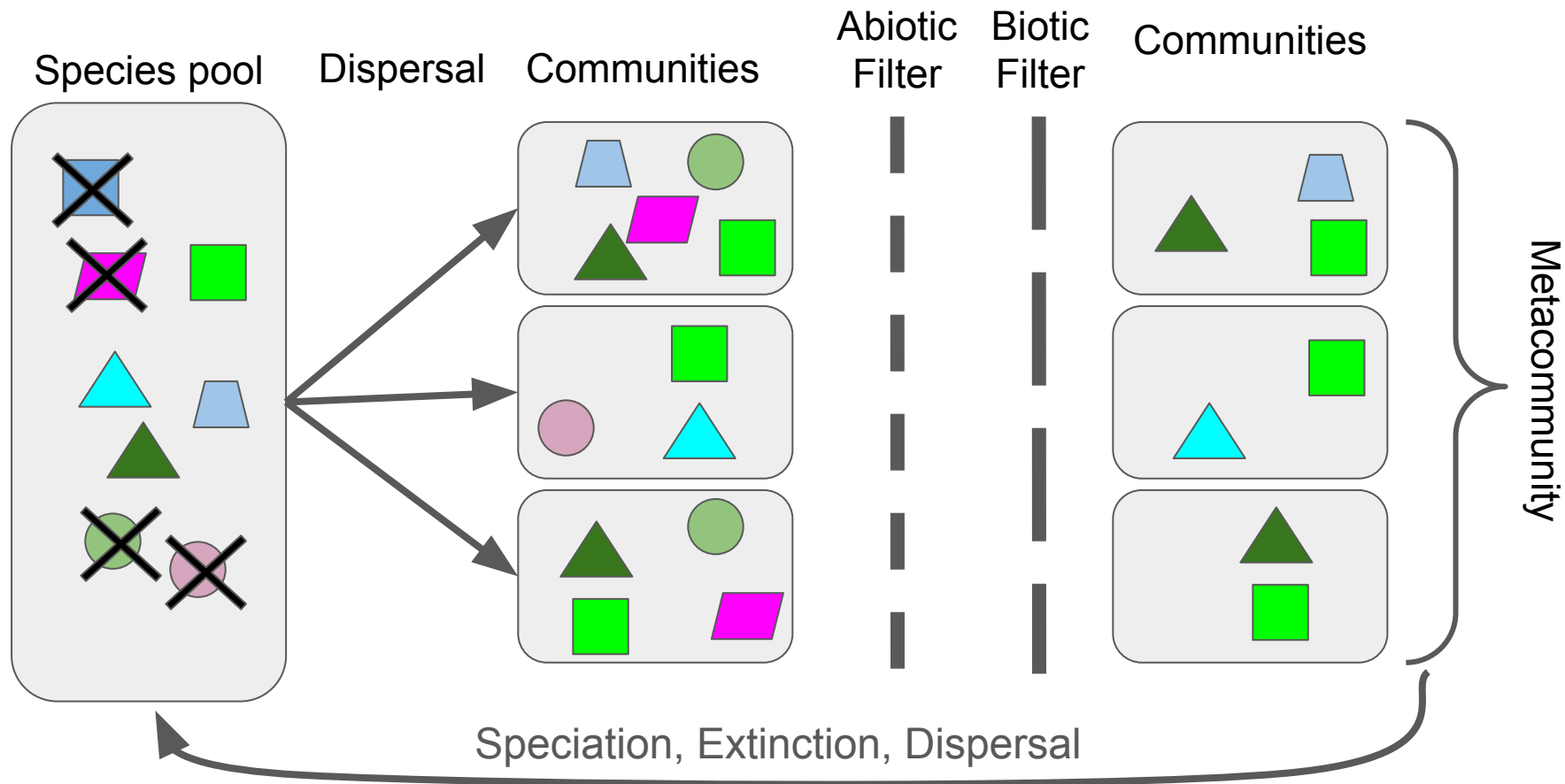












Community assembly

Community assembly focuses on a few key concepts:

- 1) Regional species pool
- 2) Dispersal
- 3) Environmental filtering
- 4) Biotic filtering

Community assembly

Community assembly focuses on a few key concepts:

- 1) Regional species pool
- 2) Dispersal
- 3) Environmental filtering
- 4) Biotic filtering

Species Pool

The species that could potentially get to a region

- How might you decide on what is in a species pool?

Species Pool



What Are Species Pools and When Are They Important?

Howard V. Cornell and Susan P. Harrison

Department of Environmental Science and Policy, University of California, Davis, California 95616; email: hvcornell@ucdavis.edu, spharrison@ucdavis.edu

The species that could potentially get to a region

- How might you decide on what is in a species pool?
 - Biogeographic regions - e.g., ecoregions, rivers, islands,
 - Fixed distance - e.g., 100 km of a given plot
 - Dispersal abilities - e.g., all species that can disperse to the site
 - Sample/plot data - e.g., all species observed in any plot/sample

Species Pool

Why do we care?



What Are Species Pools and When Are They Important?

Howard V. Cornell and Susan P. Harrison

Department of Environmental Science and Policy, University of California, Davis, California 95616; email: hvcornel@ucdavis.edu, spharrison@ucdavis.edu

Species Pool

Why do we care?

- We need to know the potential species that can get to a location
 - To know what is being excluded, we need to know our options



What Are Species Pools and
When Are They Important?

Howard V. Cornell and Susan P. Harrison

Department of Environmental Science and Policy, University of California, Davis,
California 95616; email: hvcornel@ucdavis.edu, spharrison@ucdavis.edu

Community assembly

Community assembly focuses on a few key concepts:

- 1) Regional species pool
- 2) Dispersal
- 3) Environmental filtering
- 4) Biotic filtering

Community assembly

Community assembly focuses on a few key concepts:

- 1) Regional species pool
- 2) Dispersal
- 3) Environmental filtering
- 4) Biotic filtering

Dispersal

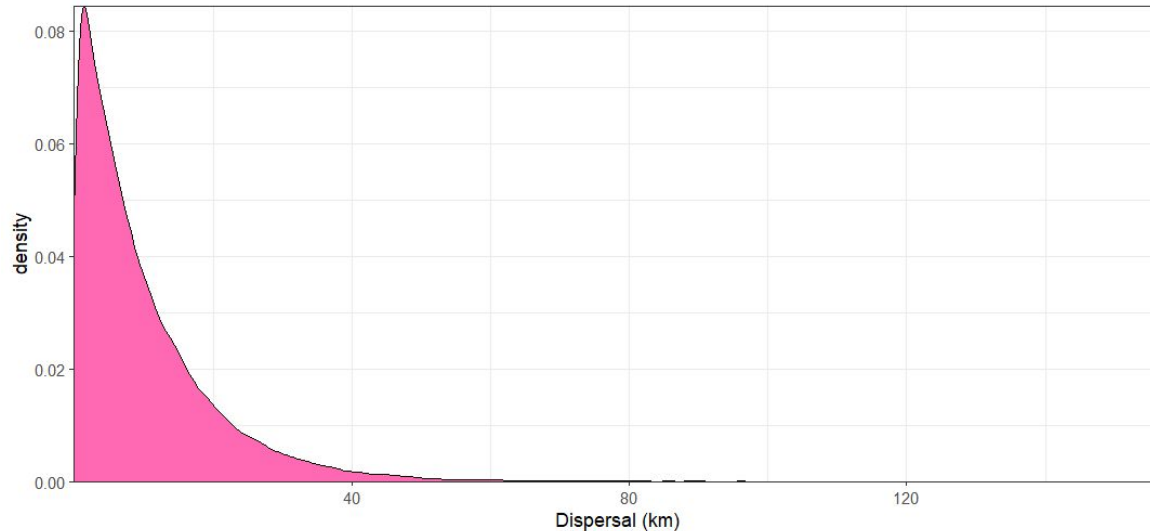
- Really hard to measure empirically
- What we really need to know is the dispersal kernel

Dispersal

- Really hard to measure empirically
- What we really need to know is the dispersal kernel
- Dispersal Kernel: distribution of dispersal distances

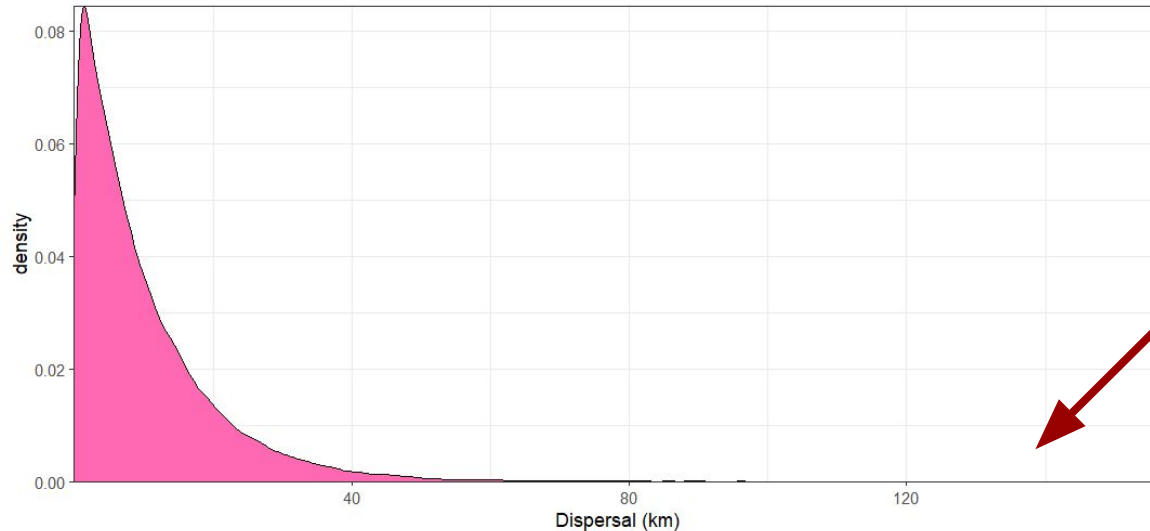
Dispersal

- Really hard to measure empirically
- What we really need to know is the dispersal kernel
- Dispersal Kernel: distribution of dispersal distances



Dispersal

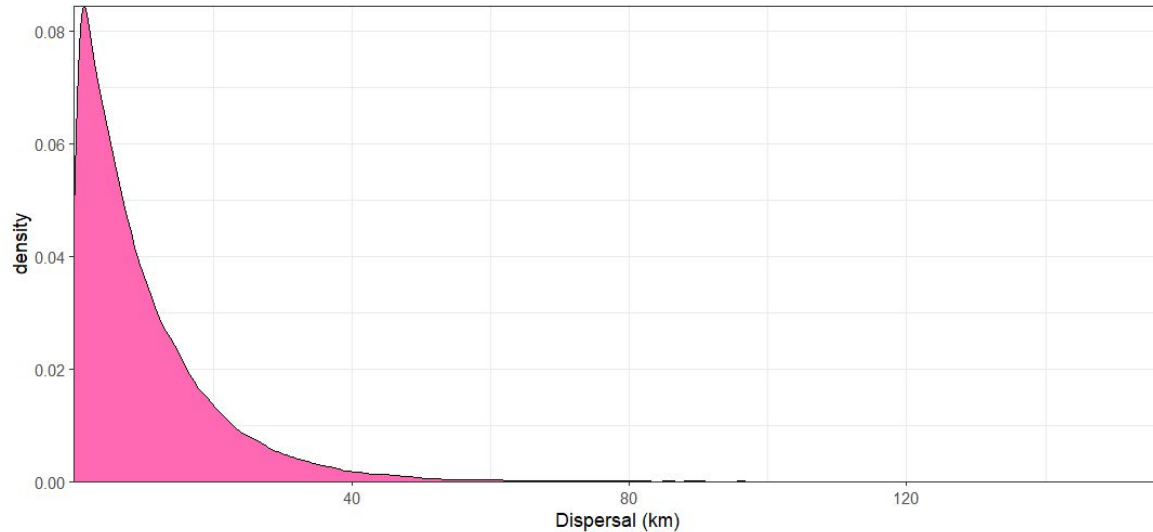
- Really hard to measure empirically
- What we really need to know is the dispersal kernel
- Dispersal Kernel: distribution of dispersal distances



These rare dispersal events are very hard to sample!

Dispersal

If long-distance dispersal events are so rare, why is it a problem?

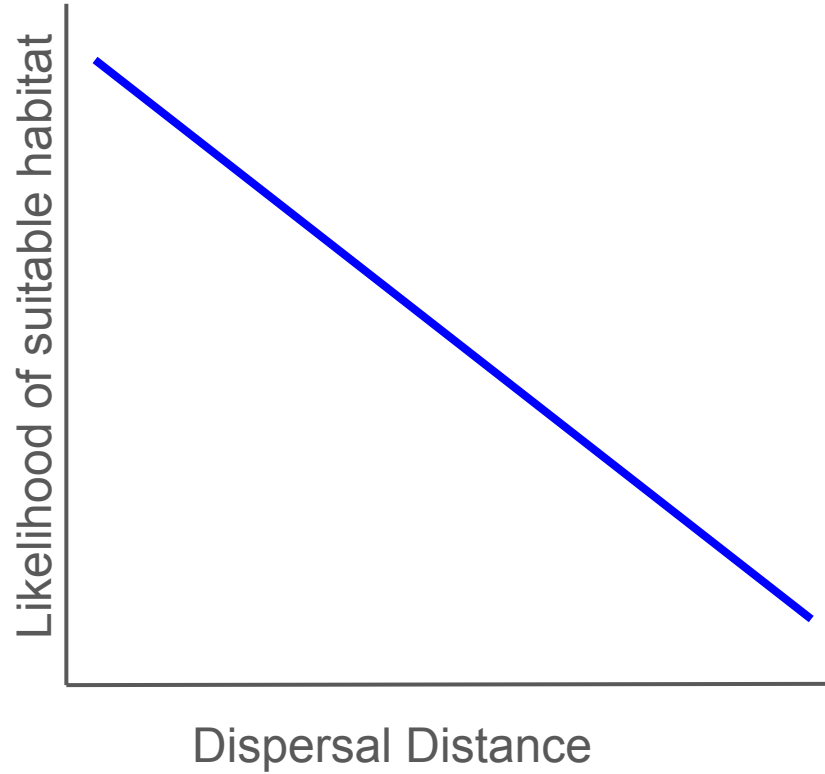


Long Distance Dispersal

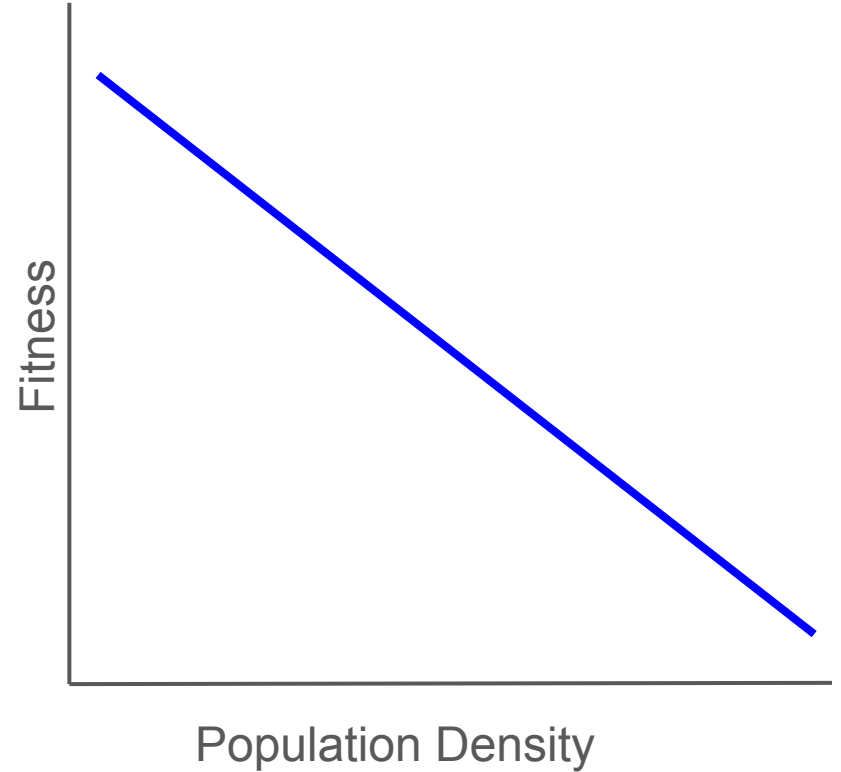
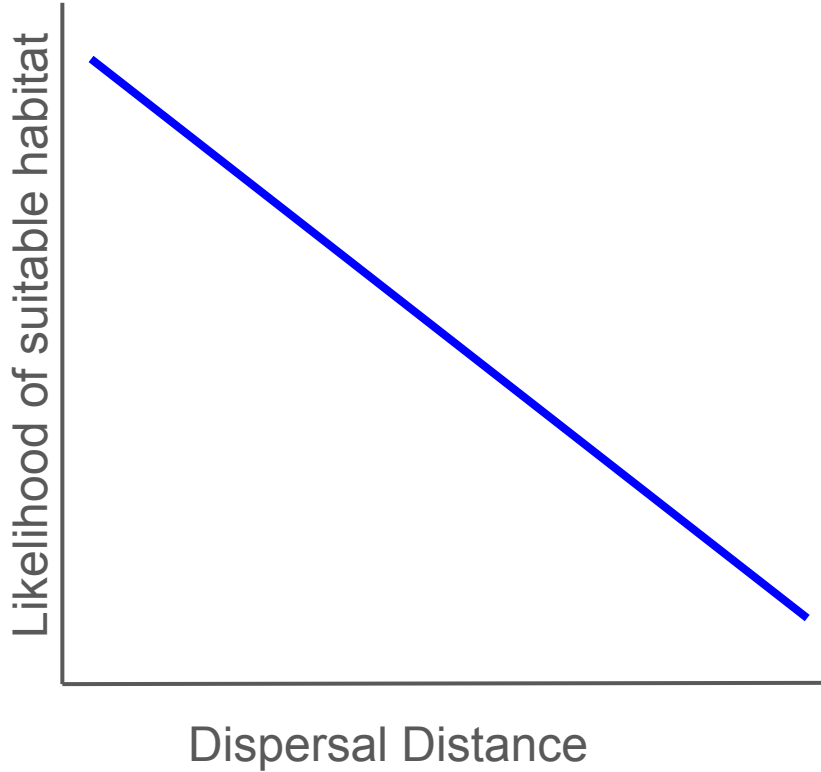
If long-distance dispersal events are so rare, why is it a problem?

- Incorrect species pools
- Long distance dispersal can be very important in dynamics
 - Rescue effects, range expansion, etc.

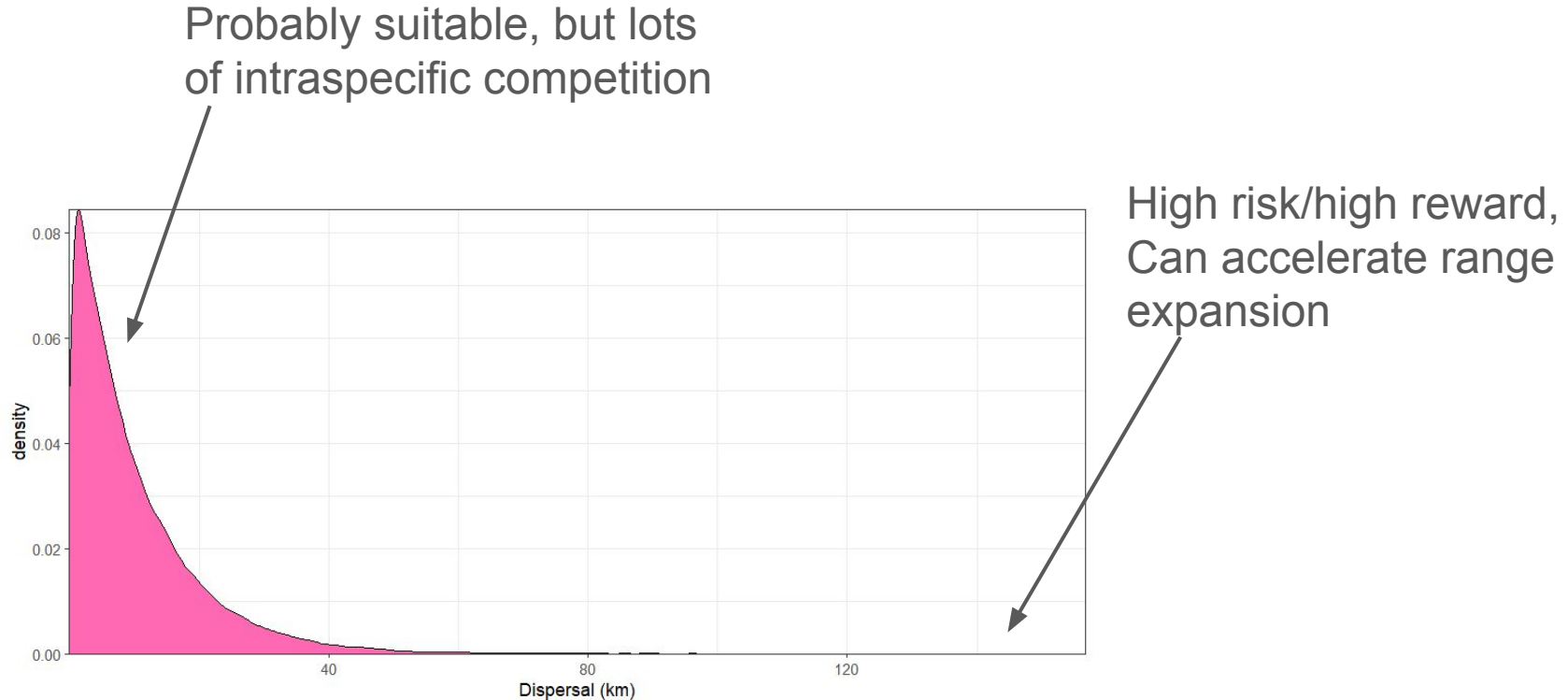
Long Distance Dispersal



Long Distance Dispersal



Long Distance Dispersal



Dispersal in practice

- We usually don't have good dispersal data
- Have to make assumptions and do the best we can
 - Most often involves assuming that anything in some spatial area or set of sites could disperse

Community assembly

Community assembly focuses on a few key concepts:

- 1) Regional species pool
- 2) Dispersal
- 3) Environmental filtering
- 4) Biotic filtering

Community assembly

Community assembly focuses on a few key concepts:

- 1) Regional species pool
- 2) Dispersal
- 3) Environmental filtering
- 4) Biotic filtering

Environmental Filtering

Can a species survive the abiotic conditions?

Environmental Filtering

Can a species survive the abiotic conditions?



Environmental Filtering

Can a species survive the abiotic conditions?



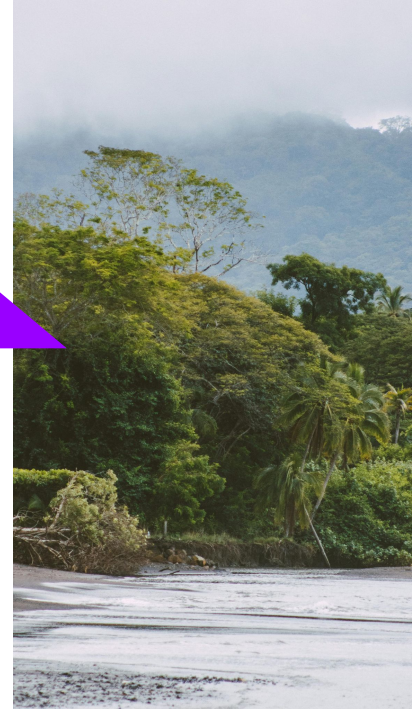
Environmental Filtering

Can a species survive the abiotic conditions?



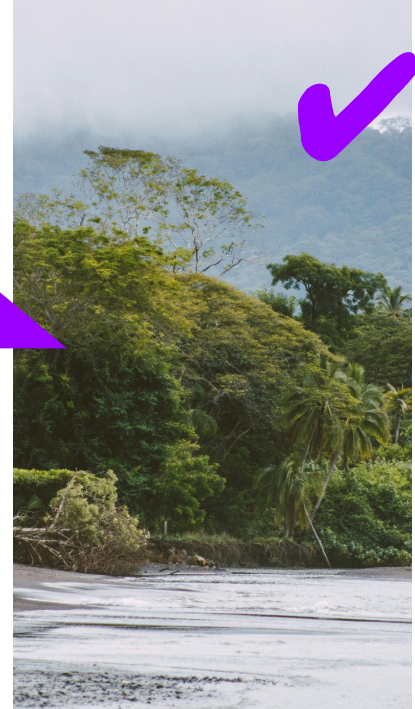
Environmental Filtering

Can a species survive the abiotic conditions?



Environmental Filtering

Can a species survive the abiotic conditions?



Environmental Filtering

Seems easy, right?

- Not always

Inferring Environmental Filtering

Multiple approaches

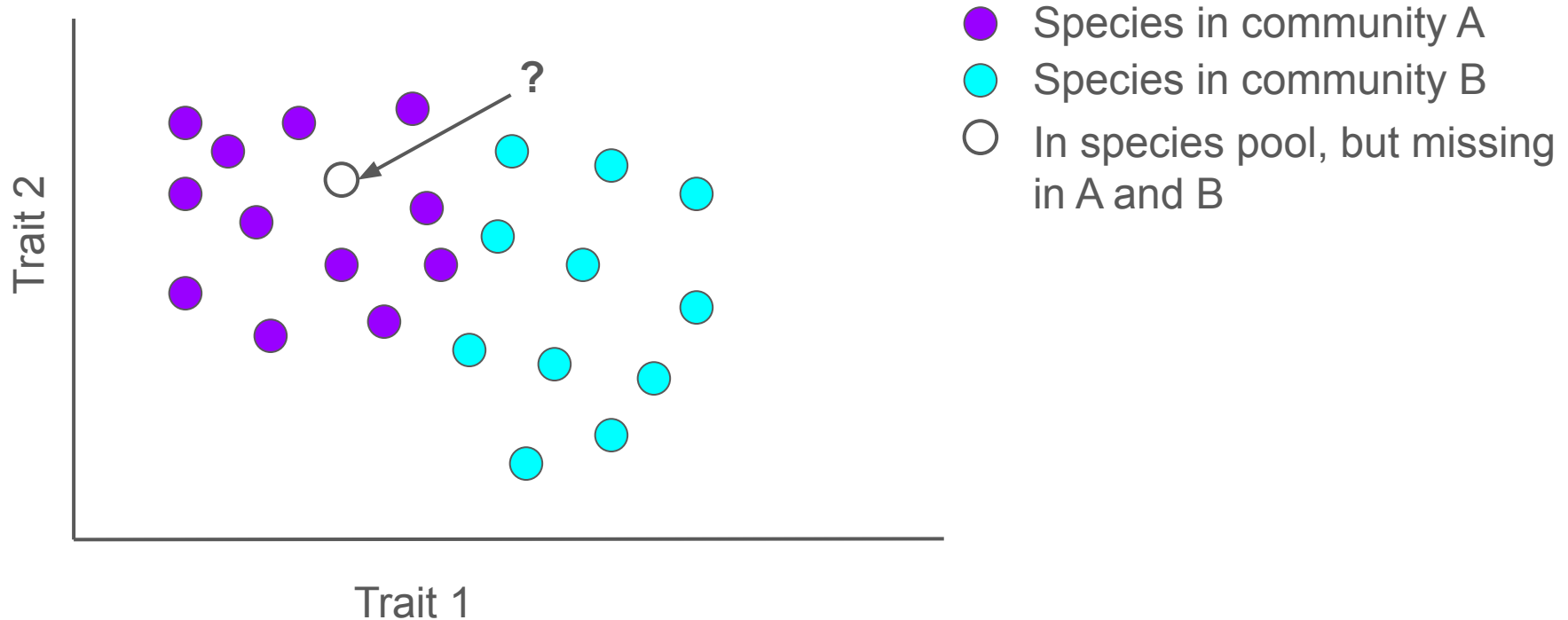
Correlative: e.g, are species similar to existing species?

Mechanistic: e.g., can the species survive there?

Experimental: e.g., can the species survive if we increase temperature?

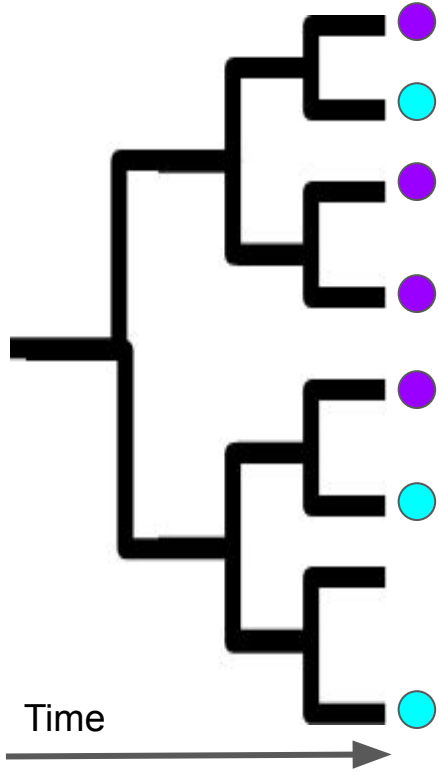
Inferring Environmental Filtering

Correlative approaches: Traits



Inferring Environmental Filtering

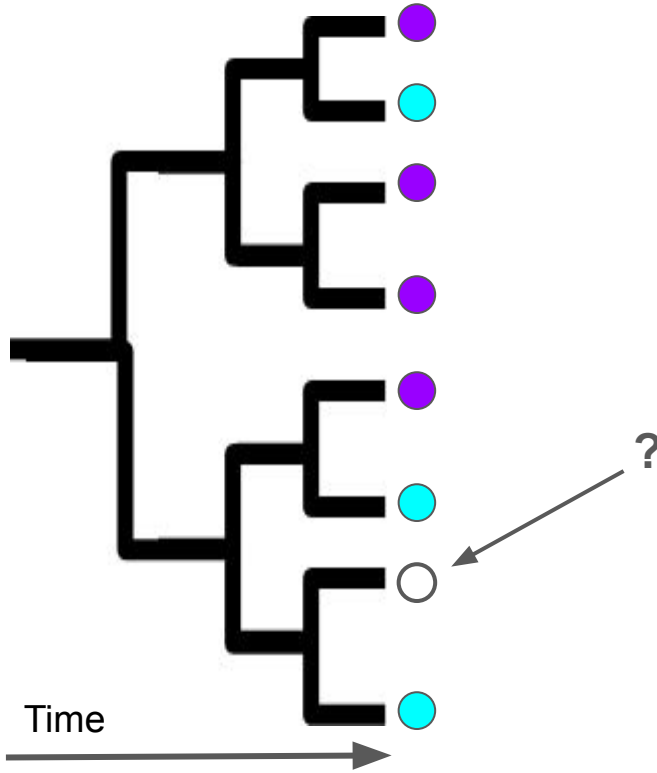
Correlative approaches: Phylogeny



- Species in community A
- Species in community B

Inferring Environmental Filtering

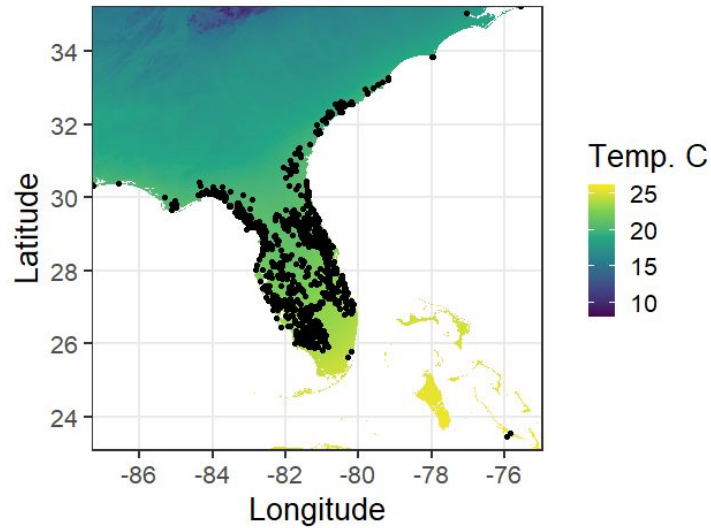
Correlative approaches: Phylogeny



- Species in community A
- Species in community B
- In species pool, but missing in A and B

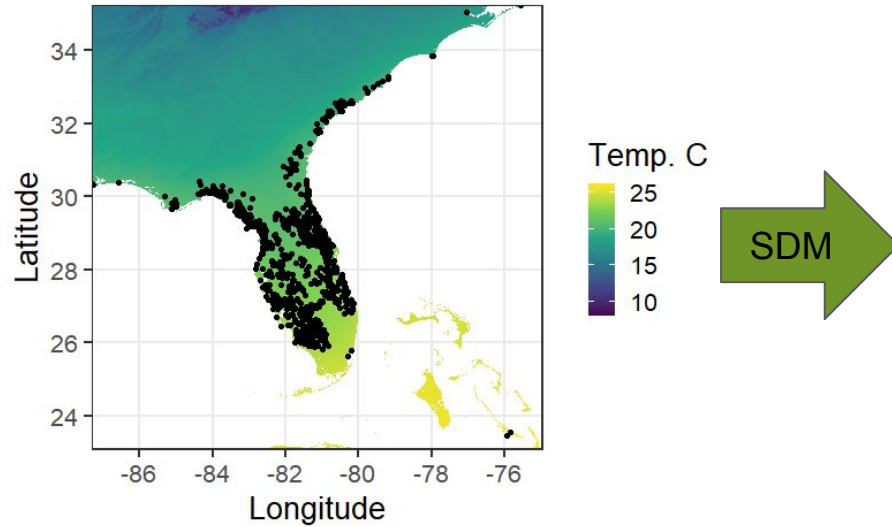
Inferring Environmental Filtering

Correlative approaches: Environmental data



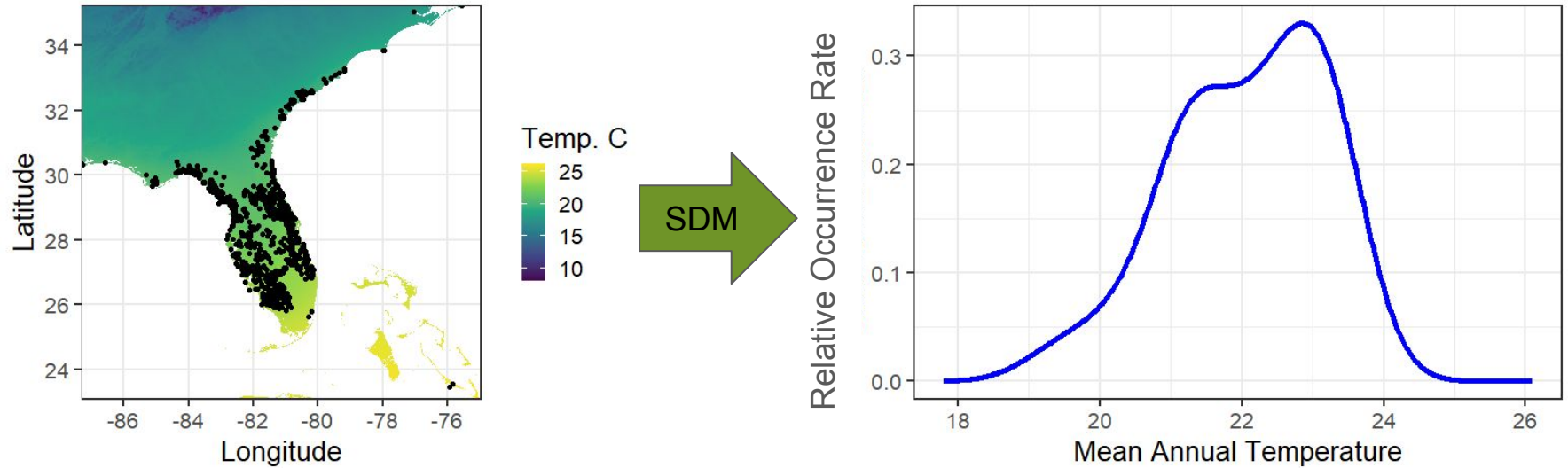
Inferring Environmental Filtering

Correlative approaches: Environmental data



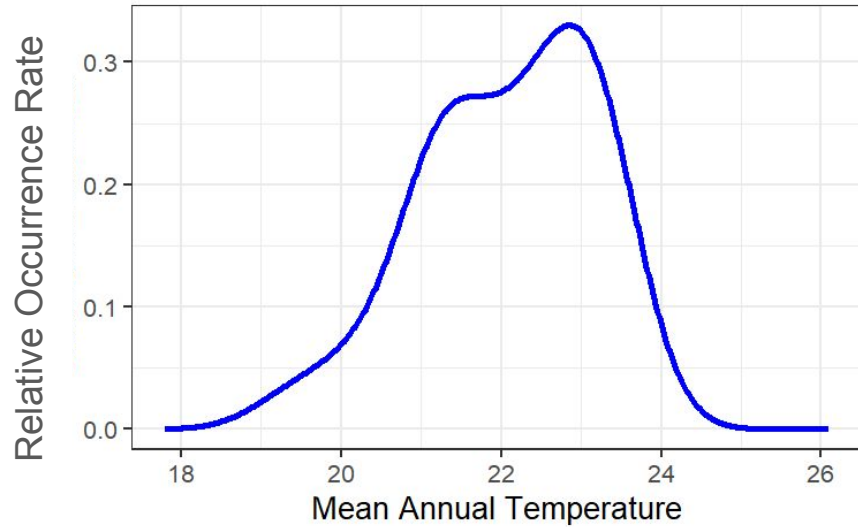
Inferring Environmental Filtering

Correlative approaches: Environmental data



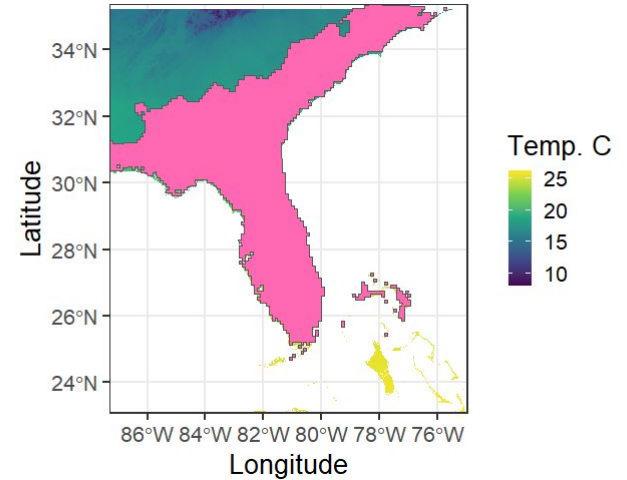
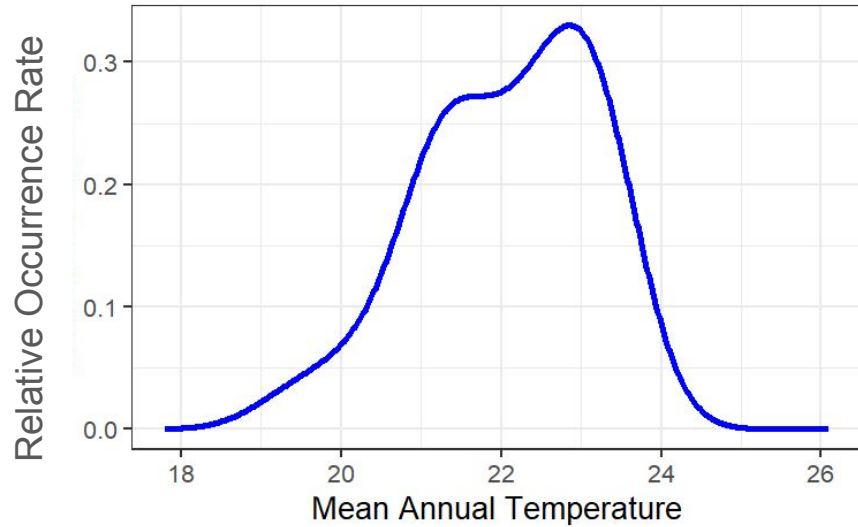
Inferring Environmental Filtering

Correlative approaches: Environmental data



Inferring Environmental Filtering

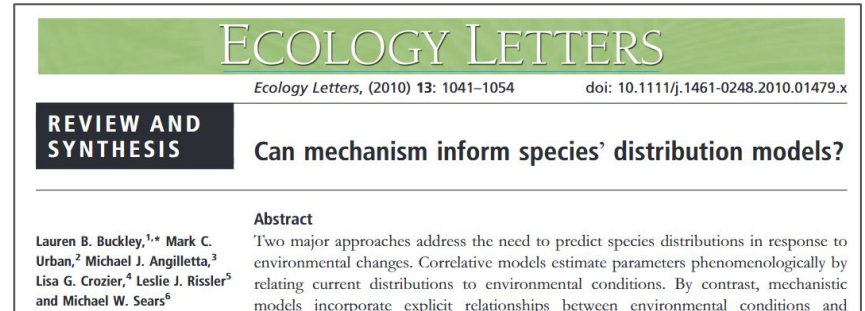
Correlative approaches: Environmental data



Inferring Environmental Filtering

Mechanistic approaches

- Measure important variables in a controlled setting
 - E.g., hatching success vs temperature, salinity vs growth rate
- Build a complicated model that is appropriate for a particular species
 - For reptiles, this might involve behavioral changes as well as physiology
 - For plants, this might involve photosynthesis temperature-response curves
- Much more data-intensive, but may not give better predictions



Inferring Environmental Filtering

Experimental approaches

- Manipulate environmental conditions and see how species respond
- E.g., warming experiments, drought experiments



Inferring Environmental Filtering

Challenges:

Species absence from an environment may not be due to a lack of suitability

Mechanistic models require a lot of data we don't have

Experiments are hard and expensive

Community assembly

Community assembly focuses on a few key concepts:

- 1) Regional species pool
- 2) Dispersal
- 3) Environmental filtering
- 4) Biotic filtering

Community assembly

Community assembly focuses on a few key concepts:

- 1) Regional species pool
- 2) Dispersal
- 3) Environmental filtering
- 4) Biotic filtering

Biotic Filtering

Can a species tolerate the biotic interactions?

- Predation/Herbivory/Parasitism/Disease
- Prey availability
- Mutualist availability
- Commensalism

Biotic Filtering

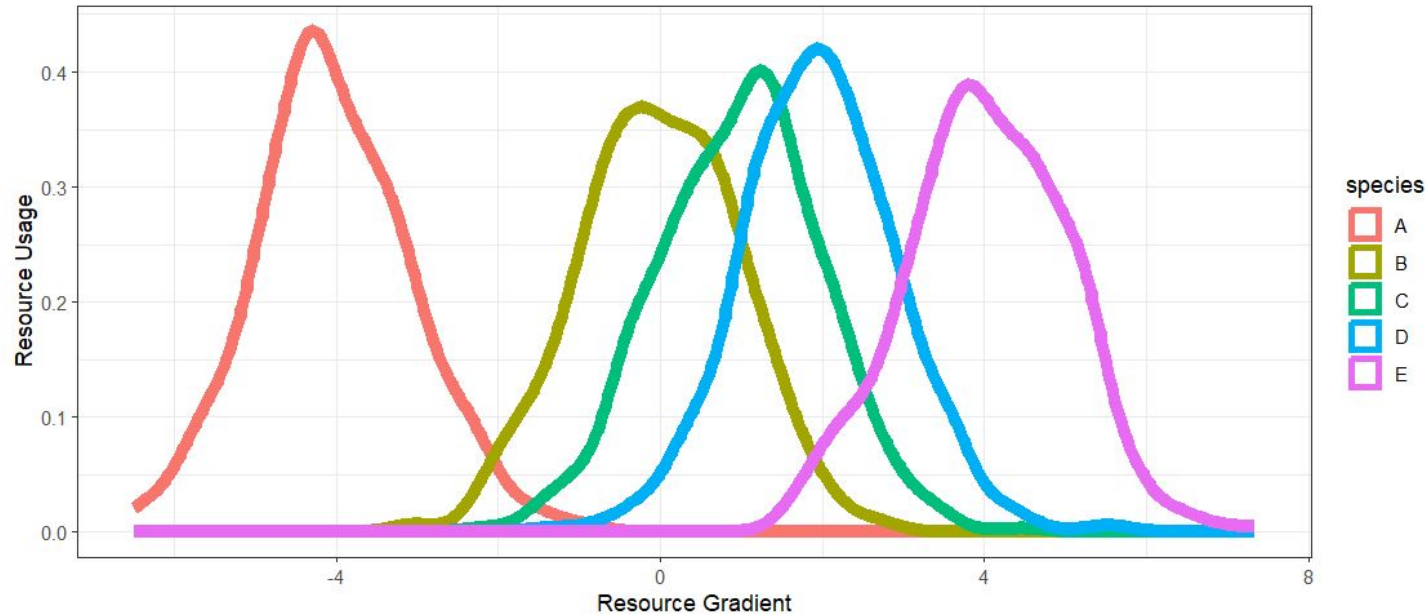
Multiple approaches

Correlative: E.g., are co-occurring species different?

Experimental: can species survive when competitors are removed?

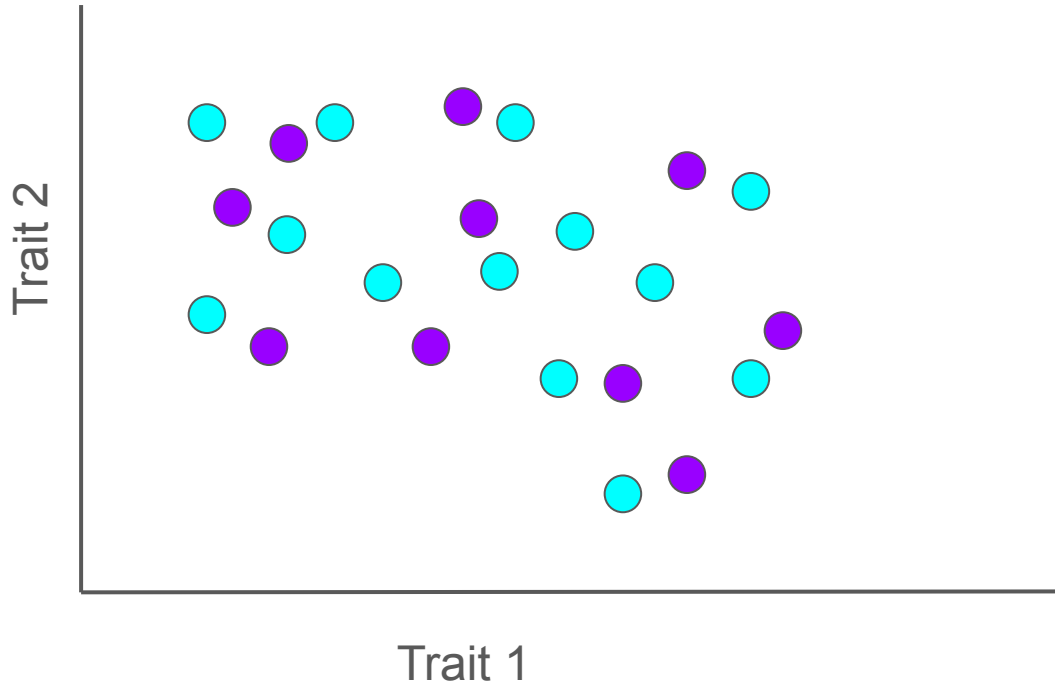
Inferring Biotic Filtering

Correlative: Are assemblages more different than you'd expect by chance?



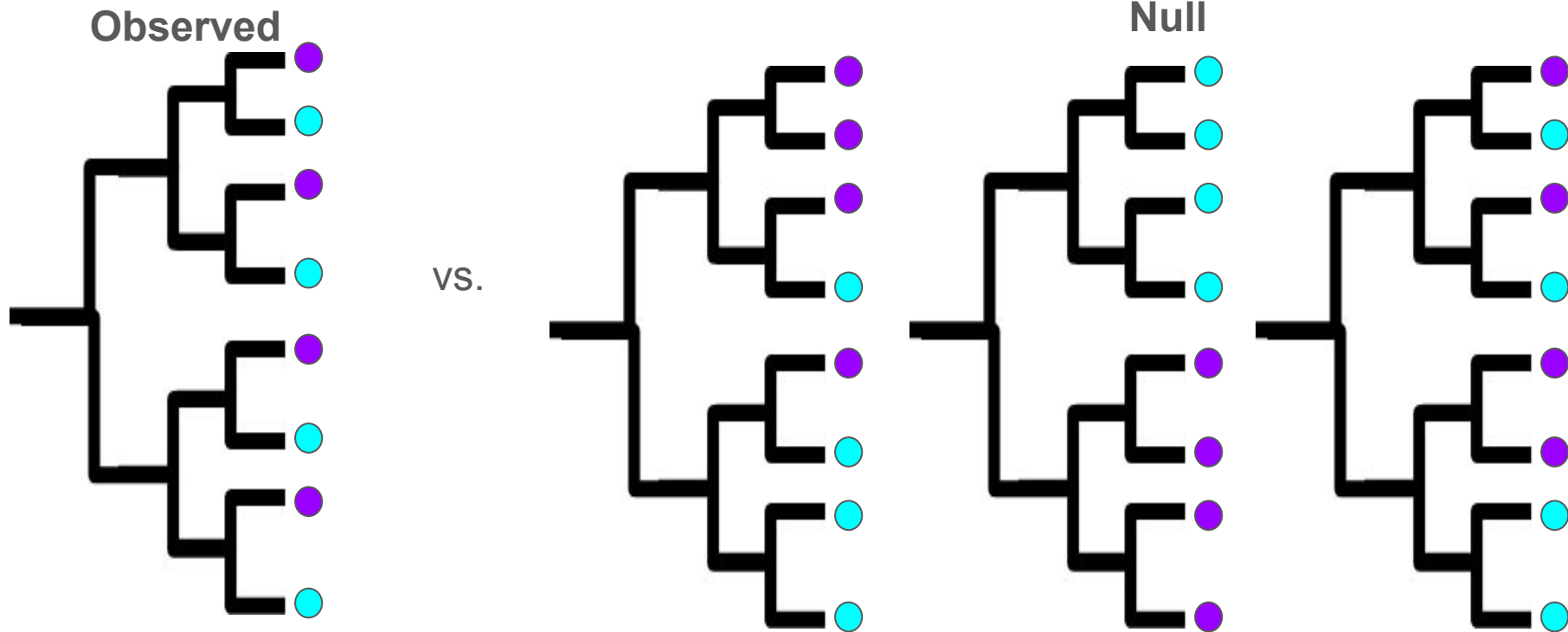
Inferring Biotic Filtering

Correlative: Are assemblages more different than you'd expect by chance?



Inferring Biotic Filtering

Correlative: Are assemblages more different than you'd expect by chance?



Inferring Biotic Filtering

Experimental: Add/remove species and see what happens

- Can a species survive in the absence of a competitor?
- Can it survive in the presence of a competitor?

Community assembly

Community assembly focuses on a few key concepts:

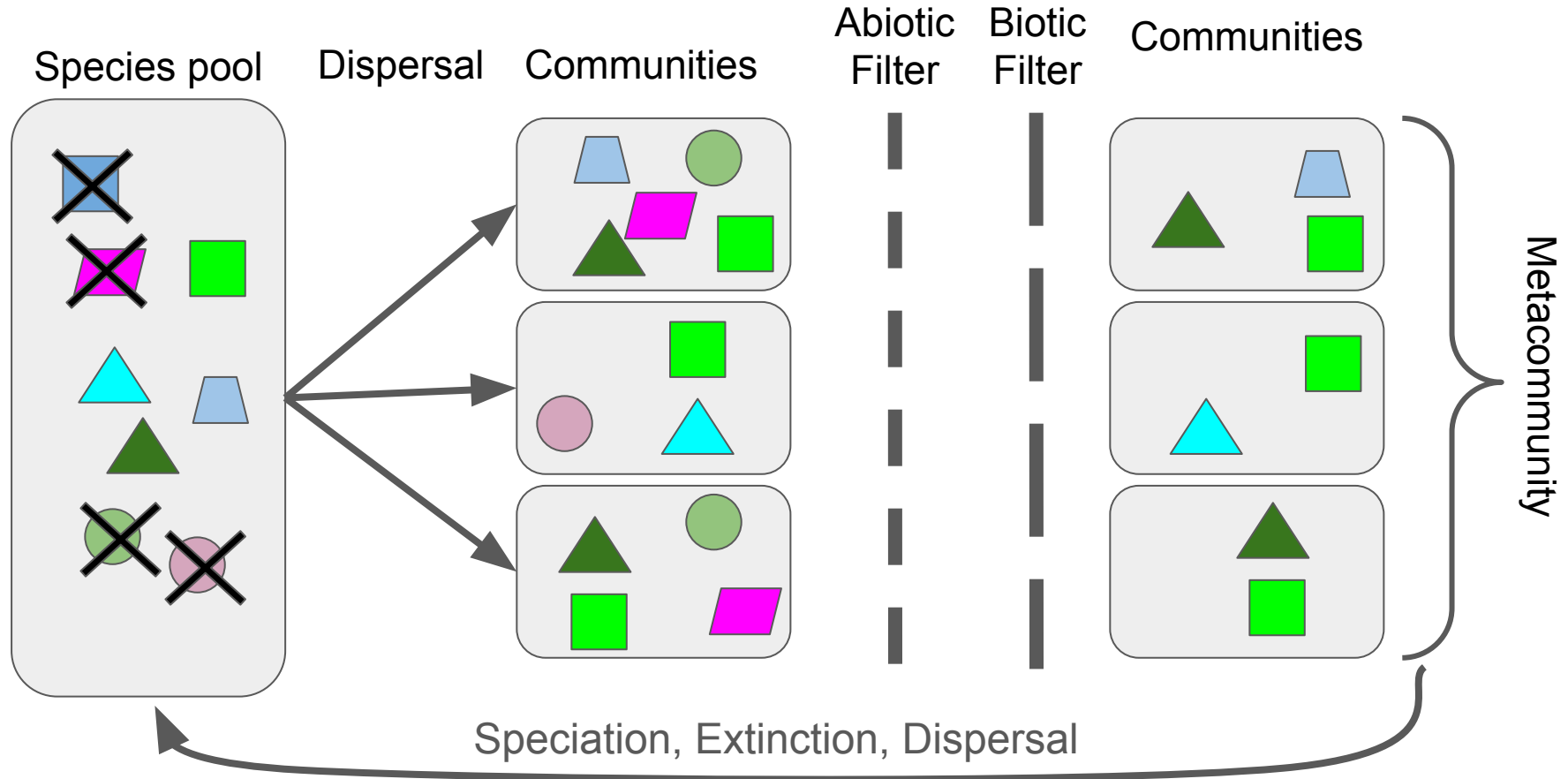
- 1) Regional species pool
- 2) Dispersal
- 3) Environmental filtering
- 4) Biotic filtering

Community assembly

Community assembly focuses on a few key concepts:

- 1) Regional species pool
- 2) Dispersal
- 3) Environmental filtering
- 4) Biotic filtering

Big Caveat: These aren't actually independent!



Lots of work to be done

- Most studies infer these processes based on pattern
- More experimental work needed
- More work integrating with theory

Next class: Community assembly examples